C **Foundations of Programming** Lab Assignments 16 – 22

A#16 – Even/Odd | A#17 – Sine Series | A#18 – Strings | A#19 – Employee | A#20 – Point

ASSIGNMENT NO. 18

String Operations in C

Perform: (i) length, (ii) reversal, (iii) equality check, (iv) palindrome check, (v) substring check.

C CODE

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s1[100], s2[100], rev[100];
    int  len, i;

    printf("Enter a string: ");
    scanf("%s", s1);

    /* i) Length */
    len = strlen(s1);
    printf("Length of string = %d\n", len);

    /* ii) Reversal */
    for (i = 0; i < len; i++)
        rev[i] = s1[len - 1 - i];
    rev[len] = '\0';
    printf("Reversed string = %s\n", rev);

    /* iv) Palindrome */
    if (strcmp(s1, rev) == 0)
        printf("The string is a Palindrome\n");
    else
        printf("The string is NOT a Palindrome\n");

    /* iii) Equality check */
    printf("Enter another string for equality check: ");
    scanf("%s", s2);
    if (strcmp(s1, s2) == 0)
        printf("Strings are equal\n");
    else
```

```
        printf("Strings are not equal\n");

    /* v) Substring check */
    if (strstr(s1, s2) != NULL)
        printf("Substring found in main string\n");
    else
        printf("Substring NOT found in main string\n");

    return 0;
}
```

SAMPLE I/O

```
Enter a string: madam
Length of string = 5
Reversed string = madam
The string is a Palindrome

Enter another string: mad
Strings are not equal
Substring found in main string
```

FUNCTIONS USED

STRLEN(S)
Count of chars before '\0'

STRCMP(A,B)
0 if equal, else non-zero

STRSTR(S,SUB)
Pointer if found, else NULL

REVERSAL
rev[i] = s1[len-1-i] in loop

FLOWCHART

Syntax error in text
mermaid version 10.9.1